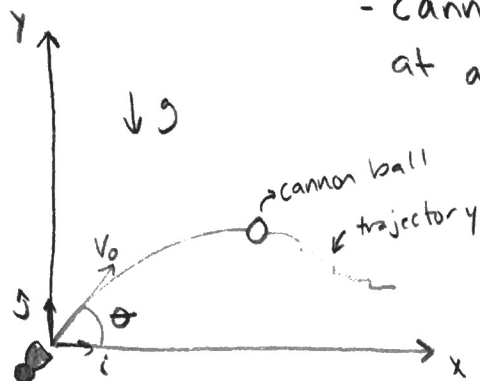


Problem 5



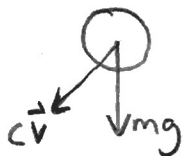
- Cannonball of mass m launched at angle θ and speed v_0 .

There is gravity and linear viscous drag.

(a) Find position vs time analytically

FBD

At arbitrary time t where cannonball is moving up and to the right.



$$m\vec{a} = -c\vec{v} - mg\hat{j} \quad \vec{v} = \dot{x}\hat{i} + \dot{y}\hat{j}$$

$$\vec{a} = \ddot{x}\hat{i} + \ddot{y}\hat{j}$$

$$m(\ddot{x}\hat{i} + \ddot{y}\hat{j}) = -c(\dot{x}\hat{i} + \dot{y}\hat{j}) - mg\hat{j}$$

two differential equations

$$m\ddot{x} = -c\dot{x} \quad m\ddot{y} = -c\dot{y} - mg$$

$$\ddot{x} = -\frac{c}{m}\dot{x}$$

$$x(t) = -c_1 e^{-c/m t} + c_2 \quad \leftarrow \text{known solution for this differential equation}$$

$$x(0) = 0 \rightarrow c_1 = c_2$$

$$\dot{x}(0) = v_0 \cos \theta$$

c_1 and c_2 are integration constants

$$\rightarrow c_1 = \frac{m}{c} v_0 \cos \theta$$

$$x(t) = \frac{m}{c} v_0 \cos \theta \left(1 - e^{-\frac{c}{m} t} \right)$$

$$\ddot{y} = -\frac{c}{m}\dot{y} - g$$

You can use integrating factor
or just guess a solution to
differential equation. It's also
a known solution

$$\dot{y} = C_3 e^{-c/m t} - g \frac{m}{c}$$

$$y(t) = C_3 e^{-c/m t} - \frac{g m}{c} t + C_4$$

$$y(0) = 0$$

$$\dot{y}(0) = v_0 \sin \theta$$

Determine C_3 and C_4 using i.c.

$$\Rightarrow \boxed{y(t) = \frac{m}{c} (v_0 \sin \theta - g t) + \left(\frac{m}{c}\right)^2 (g - e^{-c/m t} (g + \frac{m}{2} v_0 \sin \theta))}$$

(b) Find a numerical solution using $\theta = \frac{\pi}{4}$, $v_0 = 1 \text{ m/s}$, $g = 1 \text{ m/s}^2$, $m = 1 \text{ kg}$, $c = 1 \frac{\text{kg}}{\text{s}}$

From $\ddot{x} = -\frac{c}{m}\dot{x}$ and $\ddot{y} = -\frac{c}{m}\dot{y} - g$

$$z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \end{bmatrix}$$

$$\dot{z} = \begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \\ \dot{z}_4 \end{bmatrix} = \begin{bmatrix} \dot{x} \\ -\frac{c}{m}\dot{x} \\ \dot{y} \\ -\frac{c}{m}\dot{y} - g \end{bmatrix}$$

Use ODE45 to solve numerically
see attached code and plots

Analytically

$$x(2) = 0.611410 \text{ m}$$

$$y(2) = -0.523925 \text{ m}$$

Part A and B: Numerical and Analytical solutions for projectile with gravity and linear viscous drag

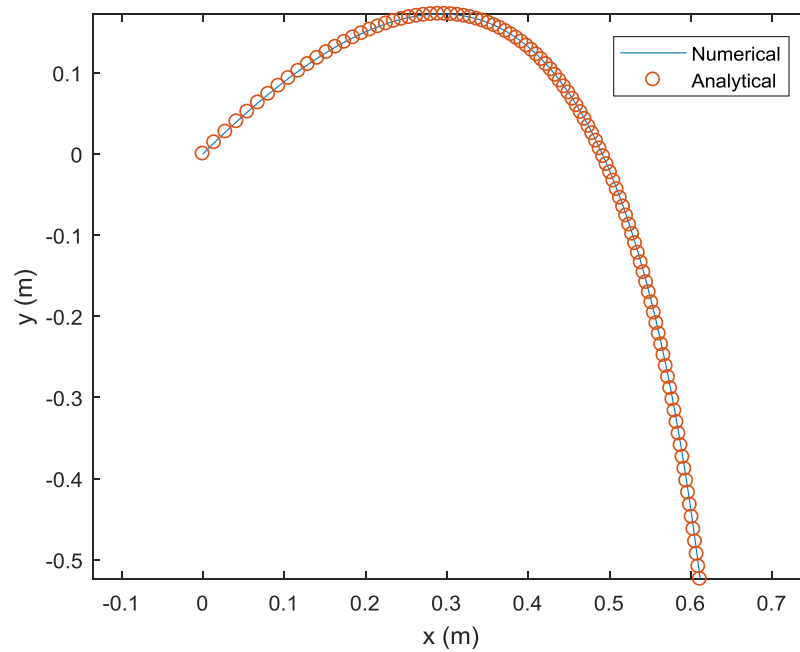


Figure 1: Trajectory of cannon ball for $t=[0,2]$ given initial condition in part C.

Part C Calculate error at $t=2$ given varying specified tolerances

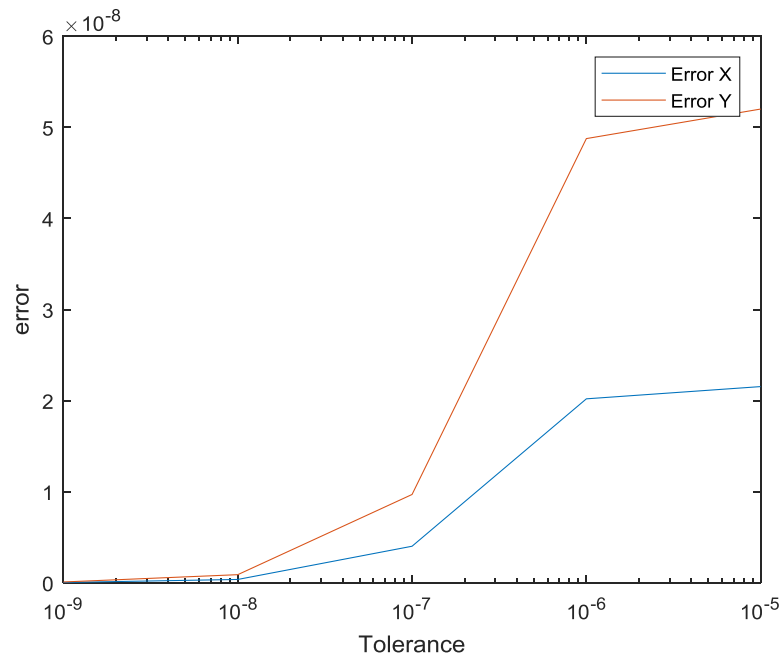


Figure 2: Error for x and y position at $t=2$ with varying 'RelTol' and 'AbsTol' tolerances.

Part D

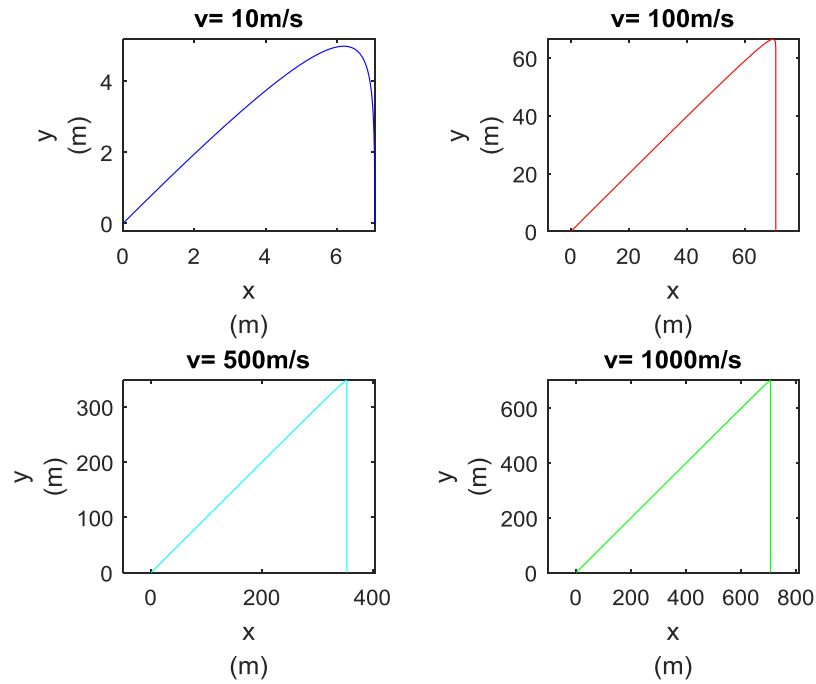


Figure 3: Plots of trajectories with increasing velocity. The of the trajectory from an asymmetric parabola goes to a triangle (linear movement with free fall)

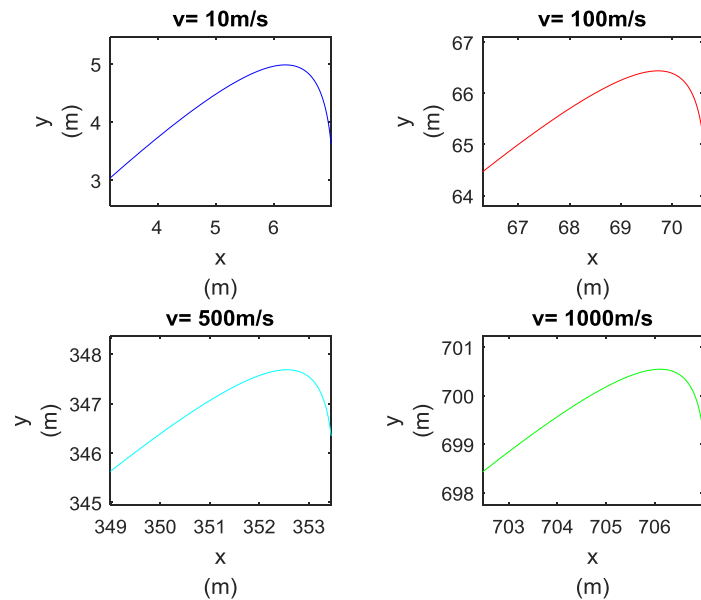


Figure 4: Plots of trajectories with increasing velocity. Zoom in at the cusp. As it can be observed the cusp is exactly the same. It just happens at different x and y positions.

Part E Calculate the best launch angle for maximizing range for a given speed. As launch speed goes to infinity, what is the best launch angle.

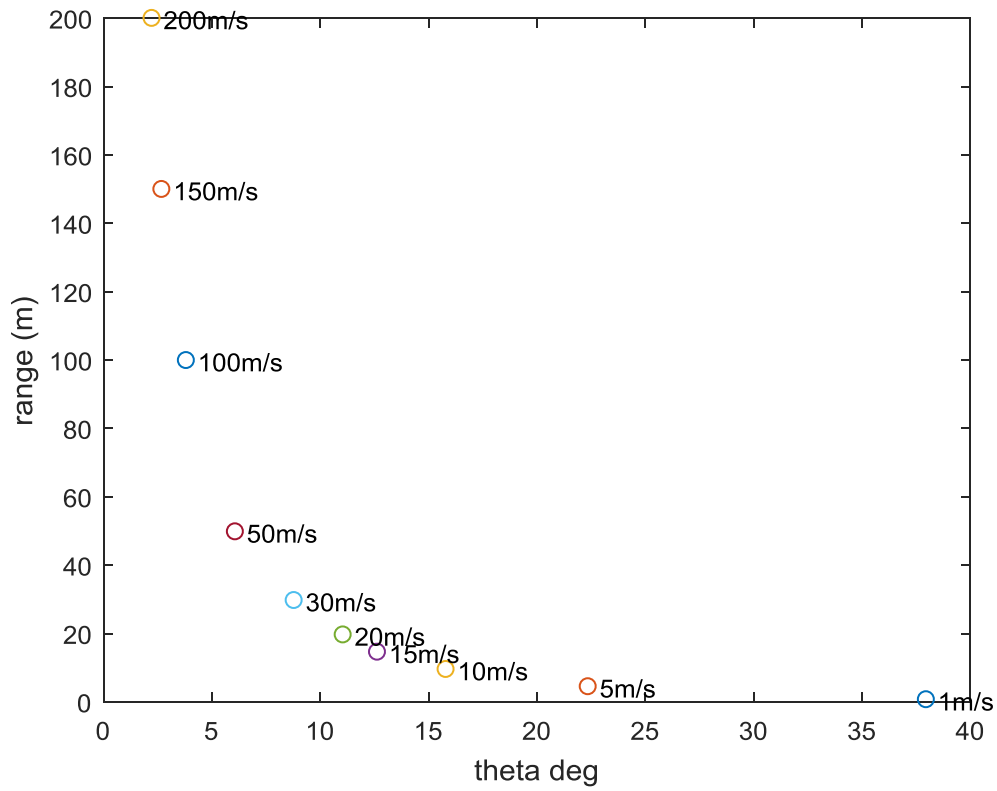


Figure 5: Plot of launch angle that yields max range for a velocity. As it can be observed as velocity goes to infinity the launch angle goes to zero.

```
%HW2 Problem 5 %cannonball with linear drag
clear all
close all
%Part A and B
%System Parameters
p.m=1; %mass kg
p.g=1; %gravity m/s^2
p.c=1; %linear drag coefficient
%initial conditions given by the problem
x_0=0; %x position at t=0
y_0=0; %y position at t=0
v0=1; %velocity at t=0
th_0=pi/4; %launch angle
xdot_0=v0*cos(th_0); %x velocity at t=0
ydot_0=v0*sin(th_0); %y velocity at t=0
inits = [ x_0 xdot_0 y_0 ydot_0]; %initial conditions state vector
t = linspace(0,2,100);
tspan=linspace(0,2,1001);
% Analytical solution
k1=p.m/p.c; k2=1/k1; %physical parameter constants to simplify code writing
xa=-k1*xdot_0*exp(-k2*t)+xdot_0*k1; %xa= x analytical
% ya=(((-p.m*ydot_0/p.c)-(p.g*((p.m/p.c)^2)))*exp(-p.c.*ta/p.m))-
% p.g*p.m.*ta/p.c+(p.m*ydot_0/p.c+p.g*((p.m/p.c)));
ya=k1*(ydot_0-p.g.*t)+((k1^2)*(p.g-(exp(-k2.*t))*(p.g+k2*ydot_0))));
% Initial Condition;
opts = odeset('RelTol',1e-9,'AbsTol',1e-9);
% Integrate!
[tarray,zarray] = ode45(@HW2_prob5_RHS,tspan,inits,opts,p); % Can use many
other integrators too (
x = zarray(:,1);
y = zarray(:,3);
%plotting
plot(x,y,xa,ya,'o')
axis equal
xlabel('x (m)')
ylabel('y (m)')
legend('Numerical','Analytical')

%Part C comparing error at t=2 with varying tolerances
n=[5 6 7 8 9];
errorx=zeros(1,length(n)); %initialize some vectors
errorx=zeros(1,length(n));
c=1;
for i=n%for loop with varying tolerances
    opts=odeset('RelTol',1/10^i,'AbsTol',1/10^i); %tolerance
    [tarray,zarray] = ode45(@HW2_prob5_RHS,tspan,inits,opts,p); % Can use
many other integrators too (
    x = zarray(:,1);
    y = zarray(:,3);
    errorx(c)=abs(x(end)-xa(end)); %error calculation
    errorx(c)=abs(y(end)-ya(end)); %error calculation
    c=c+1;
end

%plotting
figure
```

```
semilogx(1./(10.^n),errorx, 1./(10.^n), errory)
xlabel('Tolerance')
ylabel('error')
legend('Error X','Error Y')
%Part D Increasing velocity

i=1;
v1=[10 100 500 1000]; %Various increased velocities
for v=v1 %increasing velocity
    xdot_0=v*cos(th_0);
    ydot_0=v*sin(th_0);
    tspan = linspace(0, 750,10000); % initial and final times
    % Initial Condition;
    inits = [ x_0 xdot_0 y_0 ydot_0];
    opts = odeset('RelTol',1e-6,'AbsTol',1e-6);
    % Integrate!
    [tarray,zarray] = ode45(@HW2_prob5_RHS,tspan,inits,opts,p); % Can use many
    other integrators too (
    x = zarray(:,1);
    y = zarray(:,3);
    %creating a structure to save the times and positions for each velocity
    s(i).times=tarray;
    s(i).x=x;
    s(i).y=y;
    i=i+1;
end
%Plotting total trajectory
color='brcg';
figure
for i=1:4
    subplot (2,2,i)
    plot(s(i).x(s(i).y>-0.05), s(i).y(s(i).y>-0.05),color(i)) %only plotting
    until ground is hit
    axis equal
    xlabel({'x','(m)'})
    ylabel({'y','(m)'})
    title(['v= ',num2str(v1(i)), 'm/s'])
end
figure
%plotting zoom in area of cusp
for i=1:4
    subplot (2,2,i)
    [maxy, index]=max(s(i).y); %finding the index position of the maximum
    vertical pos
    plot(s(i).x(index-20:index+30), s(i).y(index-20:index+30),color(i)) %plotting
    around the max only
    axis equal
    xlabel({'x','(m)'})
    ylabel({'y','(m)'})
    title(['v= ',num2str(v1(i)) 'm/s'])
end
%Part E best launch angle to maximize the range
max_x=[]; %empty vector that will have max range
v1=[1 5 10 15 20 30 50 100 150 200]; % various velocities to test
i=1;
theta=[]; %empty vector that will have the theta that yields max range
```

```
figure
for v=v1 %for each velocity
    for th=linspace(0,pi/4,200) %for each angle
        %calculates the trajectory
        xdot_0=v*cos(th);
        ydot_0=v*sin(th);
        tspan = linspace(0, 150,1000); % initial and final times
        % Initial Condition;
        inits = [ x_0 xdot_0 y_0 ydot_0];
        opts = odeset('RelTol',1e-6,'AbsTol',1e-6);
        % Integrate!
        [tarray,zarray] = ode45(@HW2_prob5_RHS,tspan,inits,opts,p);
        x = zarray(:,1);
        y = zarray(:,3);
        %finds when the ground is reached
        y=y(y>-0.001); %this has a little bit of error not an exact point where
y=0 is crossed
        x=x(y>-0.001);
        %finds the x position given when the ground is reached
        max_x=[max_x max(x)]; %vector with range
        theta=[theta th*180/pi]; %vector with the corresponding theta
    end
    [max_x2 ,ind]=max(max_x); %finds the max x range
    maxth=theta(ind); %corresponds the th value to the max range
    plot(maxth,max_x2,'o') %plot
    text(maxth+0.5,max_x2,[num2str(v) 'm/s']) %labels each point
    hold on
end
xlabel('theta deg')
ylabel('range (m)')
```